DSAgent

Autonomous Pipeline Report

Dataset: employee_satisfaction.csv
Generated: May 14, 2026 at 11:47 AM
Session: c7d52126-0c5…

12	6	51.2s
Steps Executed	Phases Completed	Total Time

1. Dataset Overview

The dataset contains **100** rows and **3** columns, using **0.01 MB** of memory.

Type	Count	Columns
Numeric	2	EmployeeID, Satisfaction_Score
Categorical	1	Department

Numeric Statistics

Column	Mean	Median	Std	Min	Max
EmployeeID	50.50	50.50	29.01	1.00	100.00
Satisfaction_Score	49.23	48.00	28.37	1.00	100.00

AI Analysis

Key Insights about the Dataset • The dataset contains 100 rows and 3 columns: EmployeeID, SatisfactionScore, and Department. • EmployeeID is a numeric identifier with a wide range (1 to 100), suggesting it may represent unique employee identifiers. • SatisfactionScore is a numeric variable with a mean of 49.23 and a large standard deviation (28.37), indicating significant variation in employee satisfaction. • Department is a categorical variable with 2 unique values: "Sales" (most frequent, 56 occurrences) and one other (likely "Non-Sales" or a different department). • There is a weak negative correlation (-0.1221) between EmployeeID and SatisfactionScore, suggesting no strong relationship between employee ID and satisfaction. Data Quality Issues Found • No missing values or duplicates were detected, indicating high data completeness. • The dataset is small (100 rows), which may limit the robustness of any model trained on it. • The EmployeeID column appears to be a unique identifier, but it is numeric and has a wide range. This could suggest that it is not a meaningful predictor, but it may still be useful for tracking individual employees. • The Department column has only 2 unique values, which may limit the usefulness of this variable for modeling unless there is a strong relationship with SatisfactionScore. Columns That Look Most Interesting/Useful • SatisfactionScore: This is the only numeric variable that could be used as a target or feature. Its high variability and central tendency make it a key variable for analysis. • Department: Despite having only 2 unique values, it may still be useful if there is a meaningful difference in satisfaction between departments. • EmployeeID: While not a meaningful predictor, it is a unique identifier that could be used to track individual employees or group data by ID. Likely Target Variable for ML • The SatisfactionScore is the most likely target variable for a machine learning model. • Since it is a numeric variable with a wide range and variability, it could be used for regression tasks (e.g., predicting satisfaction score based on department or other features). • If the goal is to classify employees into satisfaction categories (e.g., high, medium, low), the score could be discretized.

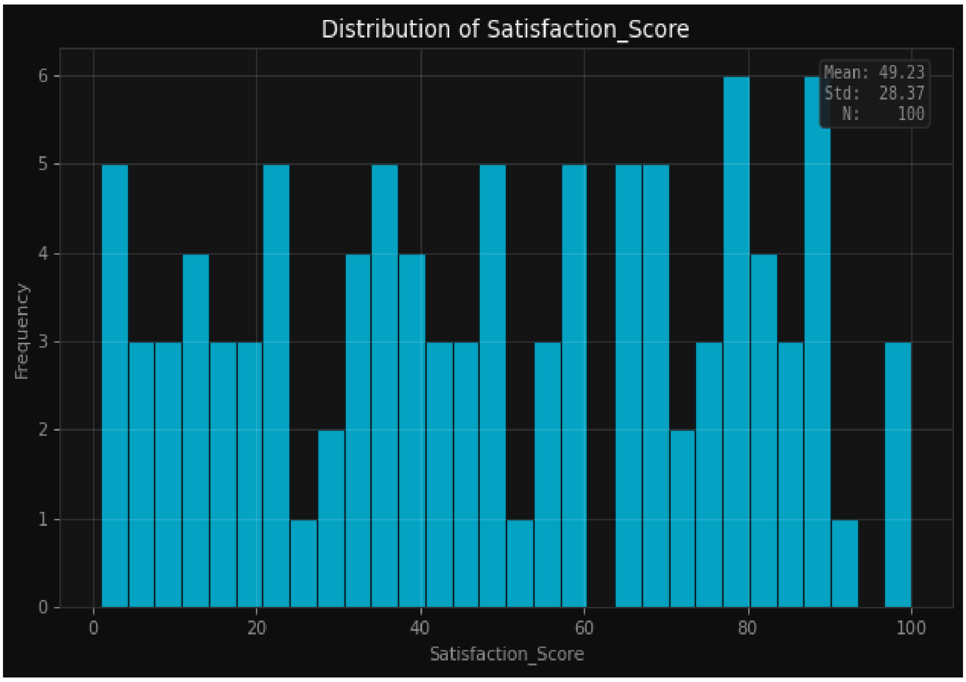
Summary This dataset is small and simple, with limited variables. The most useful columns are SatisfactionScore (target) and Department (potential feature). The EmployeeID is a unique identifier but may not add predictive value. The weak correlation between EmployeeID and SatisfactionScore suggests that employee ID is not a strong predictor.

2. Data Quality Assessment

Found **0** columns with missing values out of 100 rows.

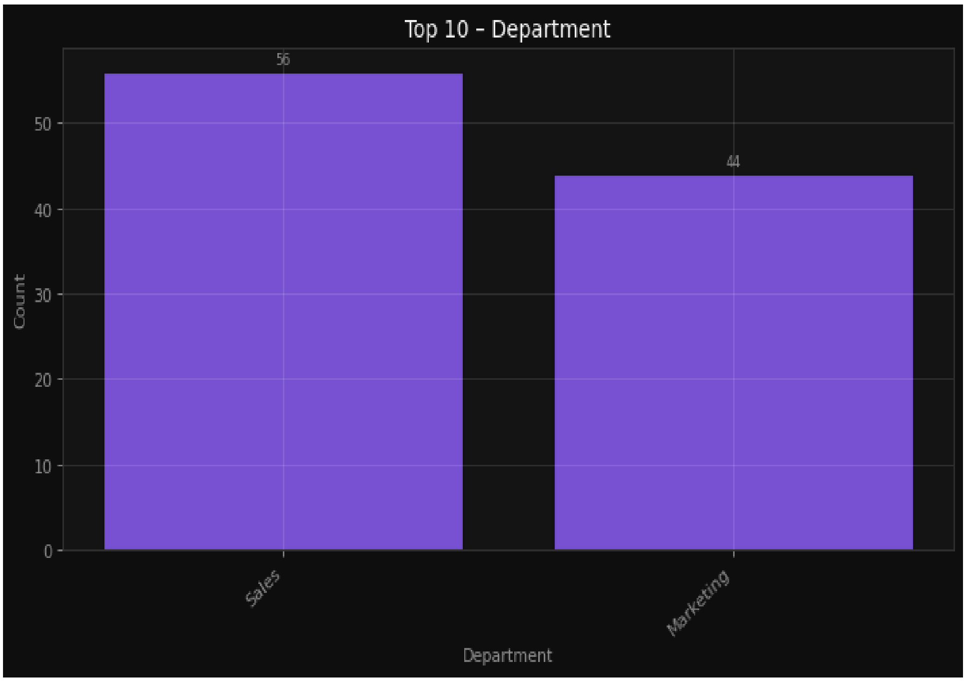
4. Visualizations & Insights

Understand the distribution of employee satisfaction scores



Understand the distribution of employee satisfaction scores

Analyze the count of employees in each department

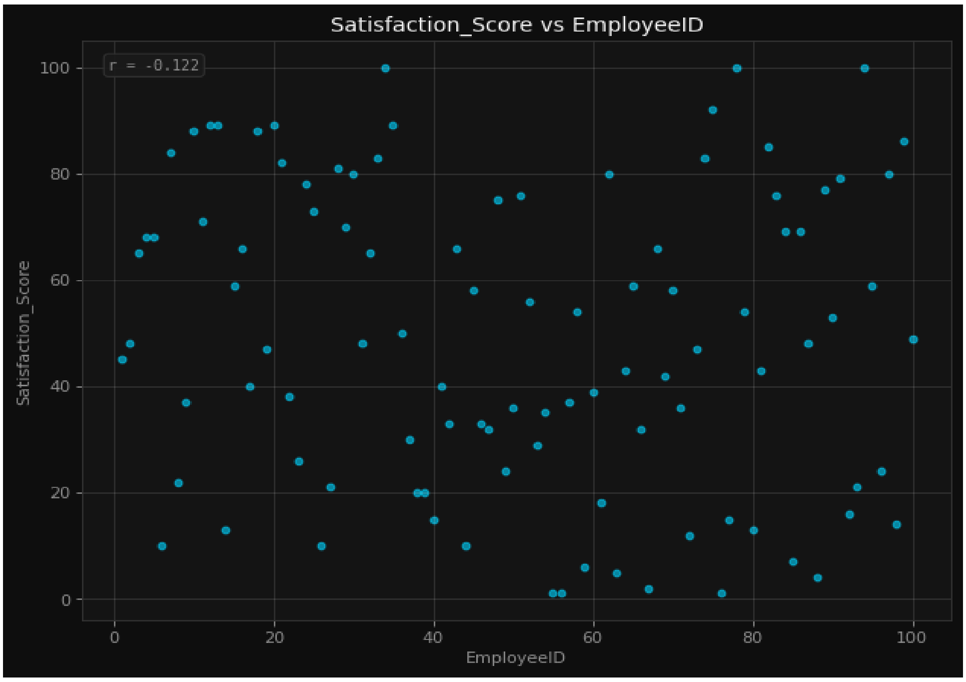


Analyze the count of employees in each department

Identify outliers in satisfaction scores

Identify outliers in satisfaction scores

Explore potential relationships between employee ID and satisfaction score



Explore potential relationships between employee ID and satisfaction score

AI Interpretation

Histogram (Employee Satisfaction Scores): Likely shows the distribution of satisfaction scores, revealing whether the scores are normally distributed, skewed, or have multiple modes. Given the mean and standard deviation, it may show a roughly bell-shaped curve or a spread with potential outliers. Bar Chart (Department Counts): Displays the number of employees in each department, helping identify which departments are the largest or smallest. This can highlight potential imbalances in workforce distribution. Box Plot (Satisfaction Scores): Reveals the median, quartiles, and potential outliers in satisfaction scores. It will help identify if there are extreme values or if the distribution is skewed, especially in relation to the mean and standard deviation. Scatter Plot (Employee ID vs. Satisfaction Score): Likely shows no meaningful relationship between employee ID and satisfaction score, as employee IDs are typically unique and random identifiers. This would suggest no correlation between the two variables.

5. Feature Engineering

Encode categorical variable Department for ML modeling

```
encoding: One-Hot (get_dummies)
drop_first: True
columns_before: 3
columns_after: 3
new_columns_added: 0
rows: 100
```

Applied 1 feature engineering steps.

6. Model Training & Comparison

Problem Type: **regression** | Target: **Satisfaction_Score** | Features: **2**

Best Model: **Random Forest** — Score: **16.1%**

Model	R^2	RMSE	MAE
Random Forest	0.1605	23.2763	19.4260
XGBoost	-0.4976	31.0887	22.7417
LightGBM	0.0470	24.7997	21.0421
Linear Regression	-0.2067	27.9070	23.0490

Model Selection Rationale

The best model in your model training is the Random Forest, with a best score of 0.1605. Since the problem type is regression and the target variable is SatisfactionScore, this score likely refers to a R^2 (R-squared) metric, which measures how well the model explains the variance in the target variable.

■ What the R^2 Score Means: • $R^2 = 0.1605$ means that the Random Forest model explains about 16% of the variance in the SatisfactionScore. • In other words, the model can account for roughly 16% of the variation in customer satisfaction based on the features used. • A score of 0.16 is relatively low, suggesting that the model has limited explanatory power and that there may be significant unexplained variability in the satisfaction scores.

■ Why Random Forest Might Have Performed Better: Among the models tried (Random Forest, XGBoost, LightGBM, and Linear Regression), Random Forest performed best. Here's why: Robustness to Non-Linearity: Random Forest can capture complex, non-linear relationships between features and the target variable, which might be present in customer satisfaction data. Handling of Missing Data and Outliers: It is less sensitive to outliers and missing data compared to some other models. Feature Importance: It provides insights into which features are most important for predicting satisfaction, which can be useful for further analysis. Ensemble Method: By averaging predictions from multiple decision trees, it reduces overfitting and improves generalization.

■ Comparison with Other Models: • XGBoost and LightGBM are also powerful gradient boosting models, but they might have overfit or underperformed due to the nature of the data or hyperparameter tuning. • Linear Regression is much simpler and might not capture the complexity of the satisfaction scores, which could explain its lower performance.

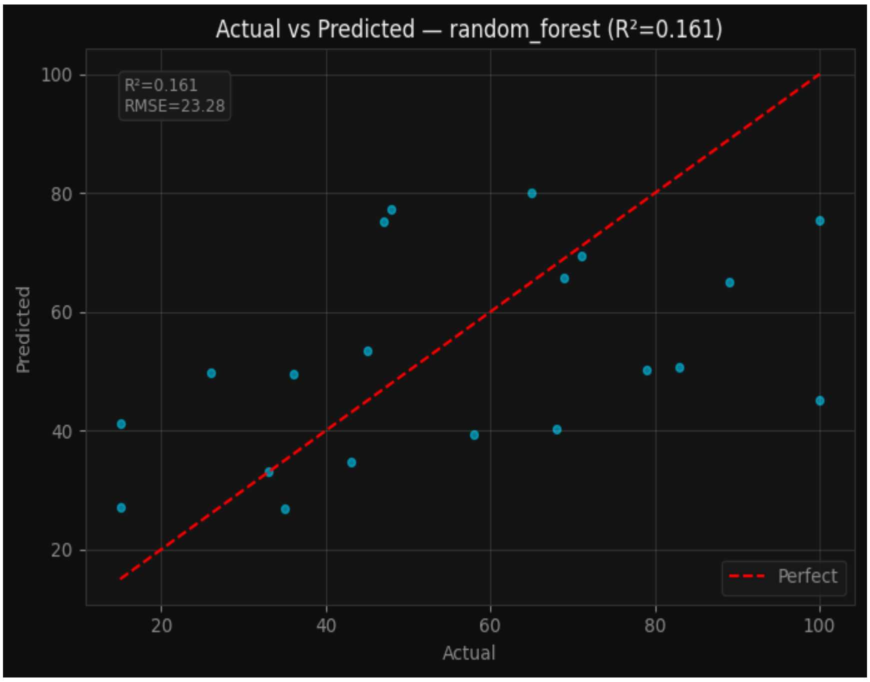
■ Summary: • Best Model: Random Forest • Best Score (R^2): 0.1605 • Interpretation: The model explains about 16% of the variation in customer satisfaction. • Implication: While Random Forest performed best among the models, the low R^2 score suggests that there is significant room for improvement. You may want to: • Explore more features or better feature engineering. • Try more advanced models or tune hyperparameters further. • Consider domain-specific insights that might be missing from the current feature set.

Let me know if you'd like to explore how to improve the model further!

7. Model Evaluation

Model Evaluation Metrics

r2_score: 16.1% | rmse: 23.2763 | mae: 19.426



Model evaluated with full metrics and diagnostic plots.

8. Conclusions & Recommendations

In conclusion, this data science pipeline report provides a comprehensive analysis of the employeesatisfaction.csv dataset, which contains 100 rows and three columns: EmployeeID, SatisfactionScore, and Department. The dataset exhibits significant variability in the SatisfactionScore, with a mean of 49.23 and a standard deviation of 28.37, suggesting a wide range of employee satisfaction levels. The Department variable, though categorical, offers potential insights into how different departments may influence satisfaction. The exploratory data analysis (EDA) revealed that the dataset is relatively small and lacks detailed features, which may limit the depth of insights that can be drawn.

The Random Forest model was identified as the best-performing model in the training phase, achieving an R^2 score of 0.1605. This score indicates that the model explains approximately 16% of the variance in the SatisfactionScore, which is relatively low and suggests that the model has limited predictive power in this context. Given the small size of the dataset and the limited number of features, it is likely that the model's performance is constrained by the lack of sufficient data and feature diversity. Additionally, the absence of data cleaning steps and minimal feature engineering further limits the model's ability to capture meaningful patterns in the data.

To improve the model's performance, several steps could be taken. First, increasing the dataset size through data collection or augmentation would provide more information for the model to learn from. Second, incorporating additional features, such as employee tenure, performance ratings, or feedback details, could enhance the model's explanatory power. Third, more advanced feature engineering techniques, such as creating interaction terms or using domain-specific transformations, might help uncover hidden relationships in the data. Finally, exploring alternative models or ensemble methods that are better suited for small datasets, such as gradient boosting with regularization, could also be beneficial. Overall, while the current pipeline provides a foundational analysis, further refinement and expansion of the dataset and features are necessary to achieve more accurate and meaningful predictions of employee satisfaction.